`PIVOT` is used to convert rows into columns.

It is particularly useful when you want to transform data from a vertical format into a horizontal report.

Let's understand with an example using Oracle SQL.

1. Understand the problem

Suppose we have a `sales` table:

Employee	Quarter	Amount
Amit	Q1	1000
Amit	Q2	1500
Amit	Q3	1200
Rahul	Q1	2000
Rahul	Q2	2500
Rahul	Q3	1800

The data is currently in a row-oriented format.

Requirement: Convert quarters into columns.

Expected output:

Employee	Q1	Q2	Q3
Amit	1000	1500	1200
Rahul	2000	2500	1800

This is exactly what `PIVOT` does.

2. Oracle PIVOT syntax

```
SELECT *
FROM table_name
PIVOT (
   aggregate_function(value_column)
   FOR pivot_column
   IN (value1, value2, value3)
);
```

Let's break it down:

Component	Purpose
`aggregate_function`	SUM, COUNT, AVG, MAX, MIN
`value_column`	Column whose values you want to aggregate

`pivot_column`	Column whose values become new columns
`IN`	Specifies which values become columns

3. Complete Oracle example

```
SELECT *
FROM sales
PIVOT (
   SUM(amount)
   FOR quarter
   IN (
      'Q1' AS Q1,
      'Q2' AS Q2,
      'Q3' AS Q3
   )
);
```

Result:

Employee	Q1	Q2	Q3
Amit	1000	1500	1200
Rahul	2000	2500	1800

Notice that:

- `quarter` values become columns.
- `amount` values are aggregated using `SUM`.
- `employee` becomes the grouping column automatically.

Important Oracle concept: Columns not used in the pivot aggregation or pivot clause become implicit grouping columns.

4. Why do we need an aggregate function?

You might wonder why we need `SUM()` when each employee has only one row per quarter.

Consider this data:

Employee	Quarter	Amount
Amit	Q1	1000
Amit	Q1	500
Amit	Q2	1500

If we execute:

```
SELECT *
FROM sales
PIVOT (
   SUM(amount)
   FOR quarter IN ('Q1' AS Q1, 'Q2' AS Q2)
);
```

Result:

Employee	Q1	Q2
Amit	1500	1500

`SUM()` aggregates multiple rows that map to the same employee and quarter.

5. PIVOT with COUNT()

Suppose you want to count the number of orders in each status.

Customer	Status
Amit	NEW
Amit	NEW
Amit	COMPLETED
Rahul	NEW
Rahul	CANCELLED

Query:

```
SELECT *
FROM orders
PIVOT (
    COUNT(status)
```

```
    FOR status IN (
        'NEW' AS new,
        'COMPLETED' AS completed,
        'CANCELLED' AS cancelled
    )
);
```

Result:

Customer	NEW	COMPLETED	CANCELLED
Amit	2	1	0
Rahul	1	0	1

6. UNPIVOT — Reverse operation

If `PIVOT` converts rows into columns, `UNPIVOT` converts columns back into rows.

Suppose we have:

Employee	Q1	Q2	Q3
Amit	1000	1500	1200
Rahul	2000	2500	1800

Query:

```
SELECT employee, quarter, amount
FROM quarterly_sales
UNPIVOT (
    amount
    FOR quarter IN (
        Q1 AS 'Q1',
        Q2 AS 'Q2',
        Q3 AS 'Q3'
    )
);
```

Result:

Employee	Quarter	Amount
Amit	Q1	1000
Amit	Q2	1500
Amit	Q3	1200
Rahul	Q1	2000
Rahul	Q2	2500

Rahul	Q3	1800

By default, Oracle `UNPIVOT` excludes rows where the unpivoted value is NULL. You can use `INCLUDE NULLS` to retain them.

7. PIVOT vs GROUP BY

GROUP BY	PIVOT
Groups rows	Converts row values into columns
Usually produces one row per group	Produces columns for specified pivot values
Uses aggregate functions	Also uses aggregate functions
Useful for summaries	Useful for cross-tab reports

For example, using `GROUP BY`:

SELECT employee, quarter, SUM(amount)

FROM sales

GROUP BY employee, quarter;

Produces:

Employee	Quarter	SUM(Amount)

Amit	Q1	1000
Amit	Q2	1500
Amit	Q3	1200

Using `PIVOT` produces Q1, Q2 and Q3 as separate columns.

8. Important interview points

1. Oracle supports `PIVOT` and `UNPIVOT` from Oracle 11g.
2. PIVOT requires an aggregate function.
3. The values in the `IN` list are explicitly specified in a normal static PIVOT.
4. Columns not involved in the pivot operation become implicit grouping columns.
5. Dynamic pivoting, where columns are not known in advance, generally requires dynamic SQL in Oracle.

9. Test your understanding

Suppose you have:

Department	Gender	Employee Count
IT	Male	10
IT	Female	5
HR	Male	6

HR	Female	4

You want this output:

Department	Male	Female
IT	10	5
HR	6	4

Which column should be used in the `FOR` clause?

Department

Gender

Employee Count

Check answer

Next useful topic: Dynamic PIVOT in Oracle, where you don't know the column values in advance.